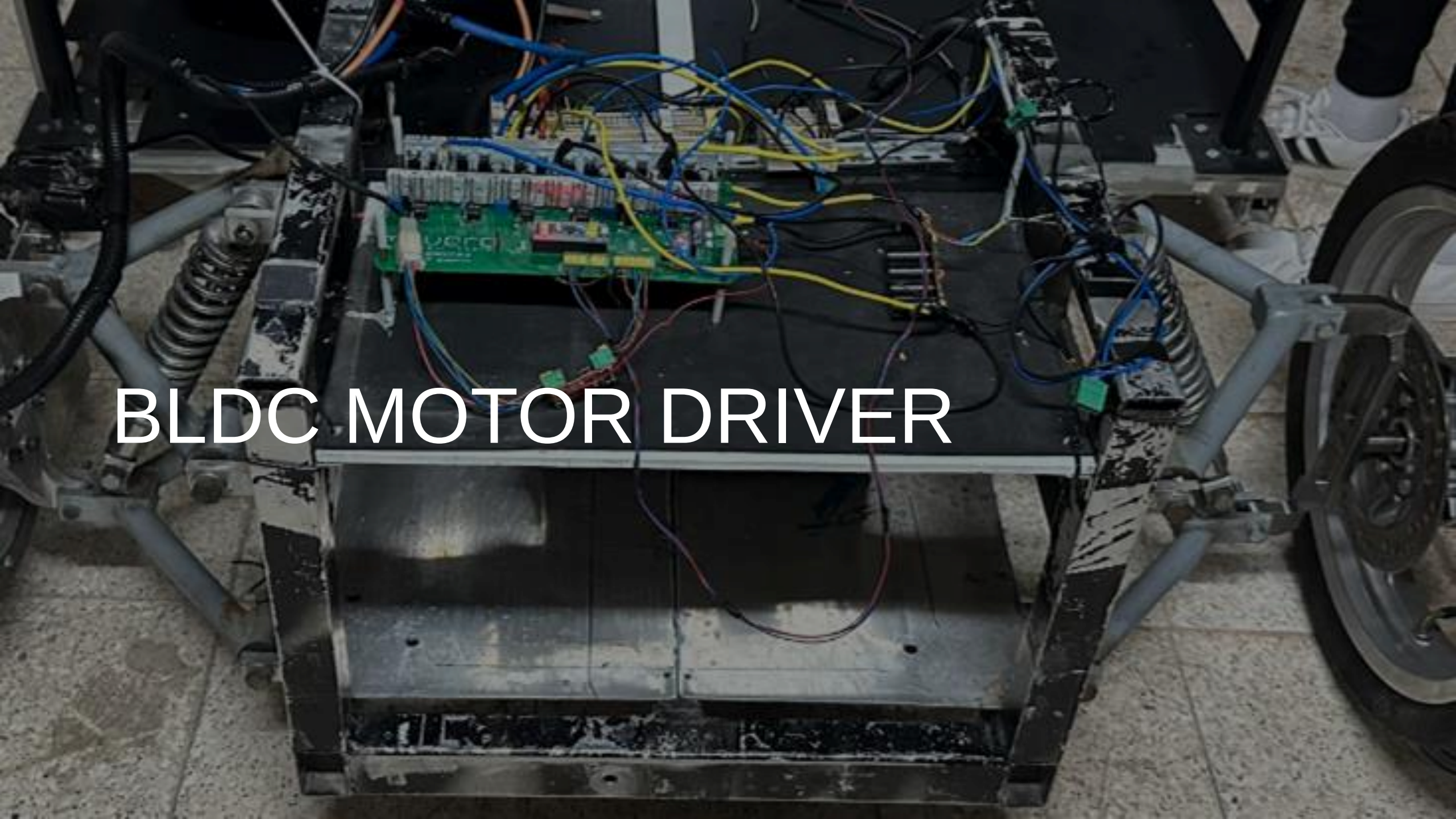

BLDC MOTOR DRIVER

Linkes

- https://youtu.be/07GMv5ZJ4MY?si=-g5qJBtcIR0HUyQt
- https://youtube.com/shorts/MVkpzZSHaQs?si=yKTzCfBzfq_a0wc2
- https://youtu.be/fH3wRFS8RlU?si=qvSBYx4WfXYujNPT
- https://youtu.be/xsq8dP8Q2FM?si=7b5MLmUhr1_sDSkU
- https://youtube.com/shorts/Nuks4e5Taco?si=nV33u2TtYmU9hOj-
- https://youtube.com/shorts/BMbomiNMHnY?si=csutMzv3415T5ybg

CONTENTS

AIM

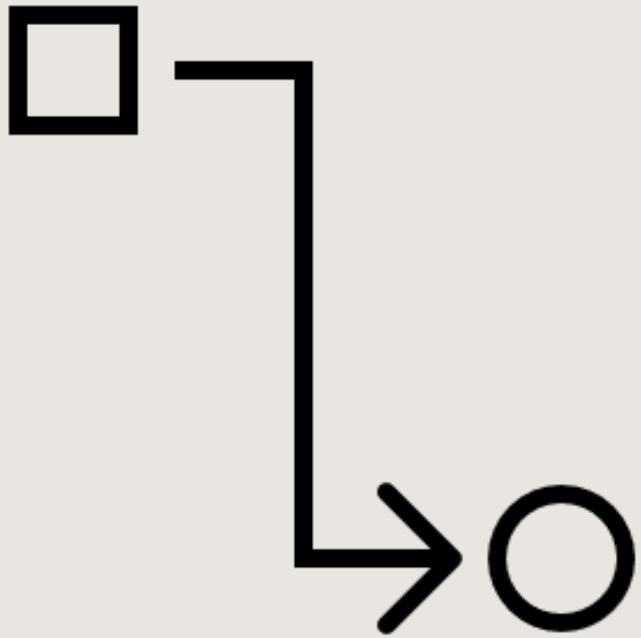
METHODOLOGY

HARDWARE

SOFTWARE

TIMING

CONCLUSION



AIM



To be a part of today's progressive electric vehicle industry.



To contribute to the motor drive industry, which is still underdeveloped in Turkey today.

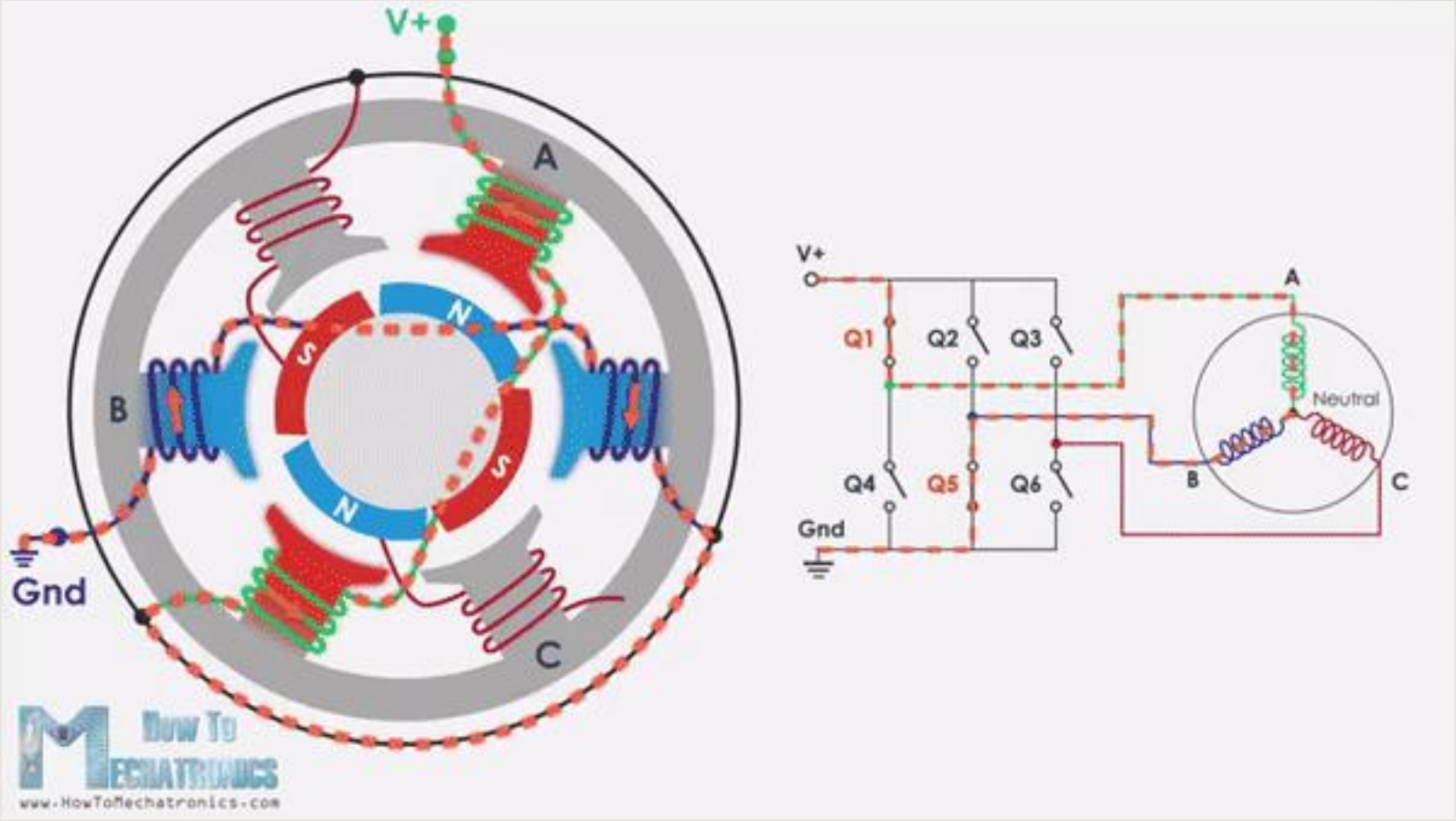


To be able to control high power circuits.



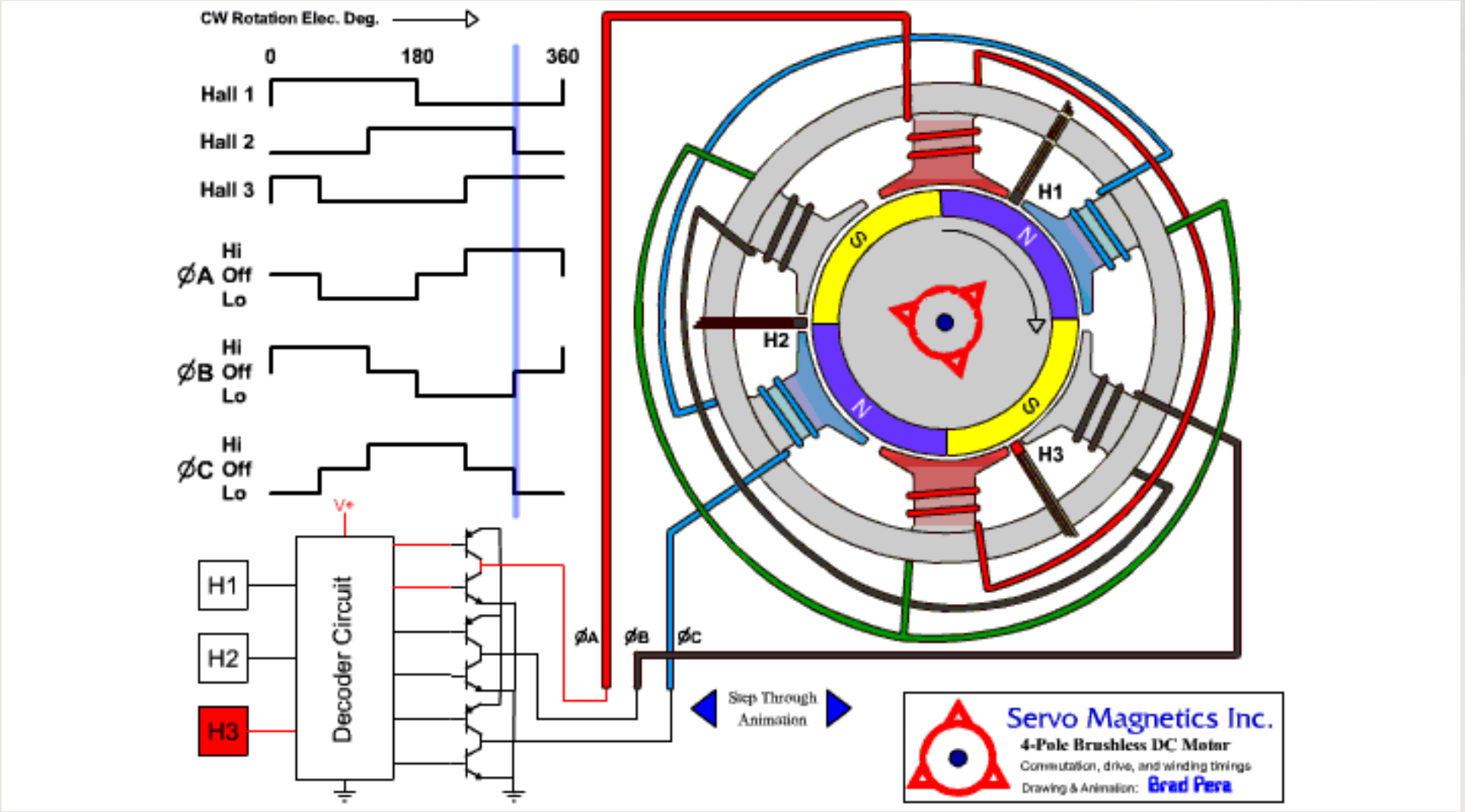
To ensure more efficient use of energy.

METHODOLOGY



[1]

BLDC motor operating logic:



[2]

120 degree BLDC motor commutation sequence:

Table 3. Clockwise direction

Commutation vector			Vector	Hall sensor pattern definition			Hall sensor pattern result
Phase A	Phase B	Phase C		Hall Sensor C	Hall Sensor B	Hall Sensor A	
NC	$+V_{DCB}$	$-V_{DCB}$	A	1	0	1	5
$-V_{DCB}$	$+V_{DCB}$	NC	B	1	0	0	4
$-V_{DCB}$	NC	$+V_{DCB}$	C	1	1	0	6
NC	$-V_{DCB}$	$+V_{DCB}$	D	0	1	0	3
$+V_{DCB}$	$-V_{DCB}$	NC	E	0	1	1	2
$+V_{DCB}$	NC	$-V_{DCB}$	F	0	0	1	1

Table 4. Counterclockwise direction

Commutation vector			Vector	Hall sensor pattern definition			Hall sensor pattern result
Phase A	Phase B	Phase C		Hall Sensor C	Hall Sensor B	Hall Sensor A	
NC	$-V_{DCB}$	$+V_{DCB}$	D	1	0	1	5
$-V_{DCB}$	NC	$+V_{DCB}$	C	0	0	1	1
$-V_{DCB}$	$+V_{DCB}$	NC	B	0	1	1	3
NC	$+V_{DCB}$	$-V_{DCB}$	A	0	1	0	2
$+V_{DCB}$	NC	$-V_{DCB}$	F	1	1	0	6
$+V_{DCB}$	$-V_{DCB}$	NC	E	1	0	0	4

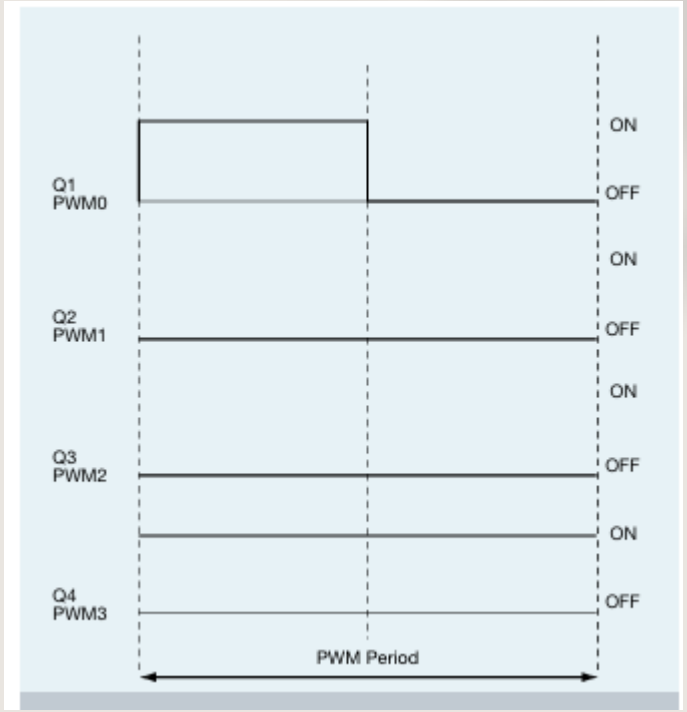
[3]

PWM modulation techniques:

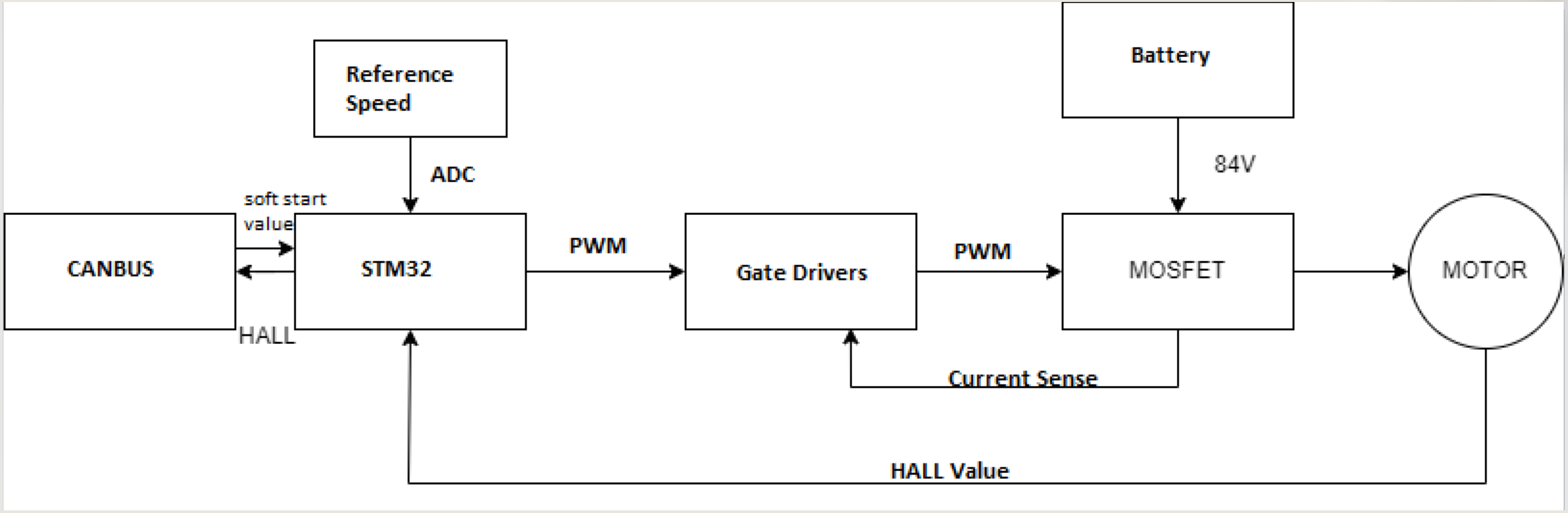
- **Independent Bipolar PWM Modulation**
- **Complementary Bipolar PWM Modulation**
- **Independent Unipolar PWM Modulation**

Independent Unipolar PWM Modulation

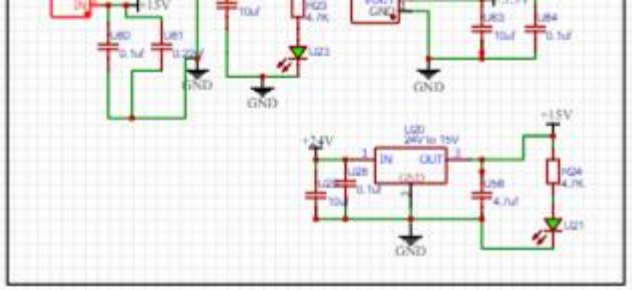
- Ease of application and control.
- Among other modulation techniques, this one puts the least stress on the mosfet.

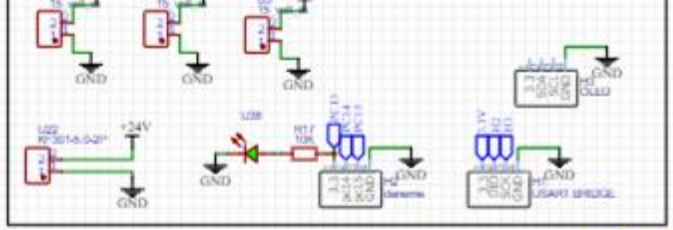


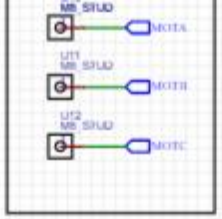
BLDC motor driver block diagram

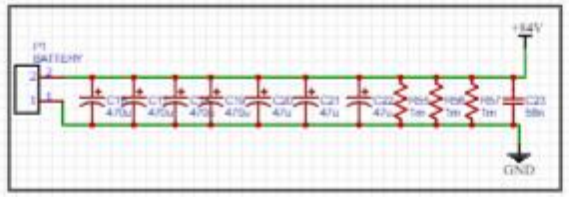


HARDWARE

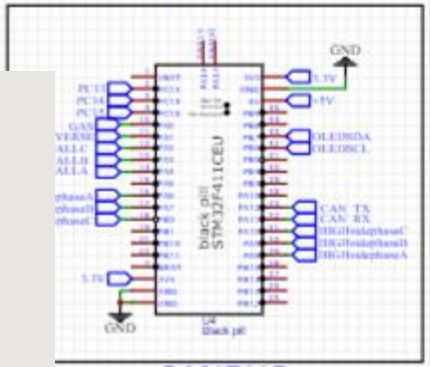




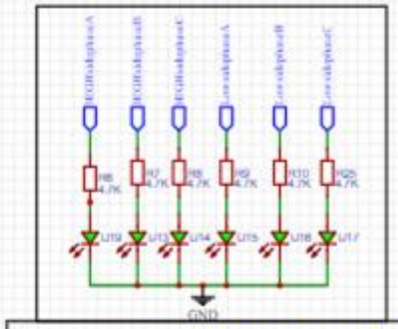




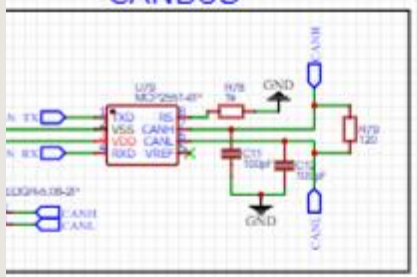
STM



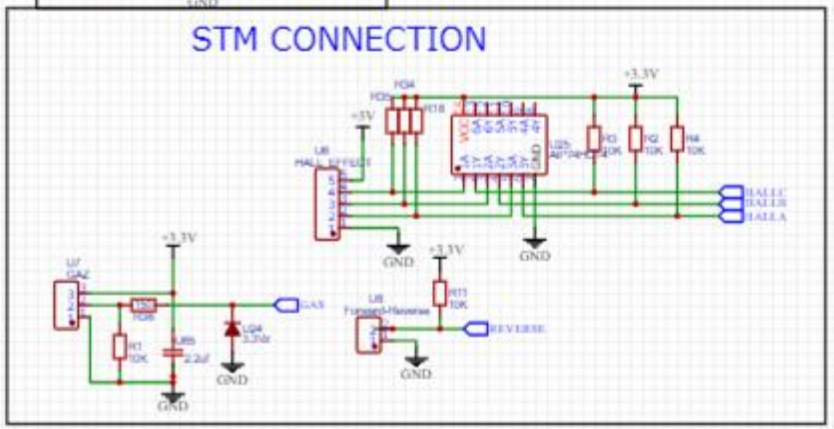
LEDS

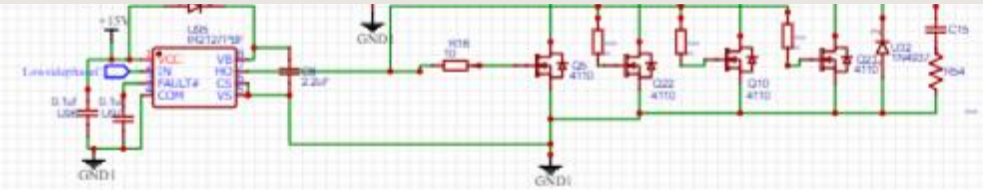


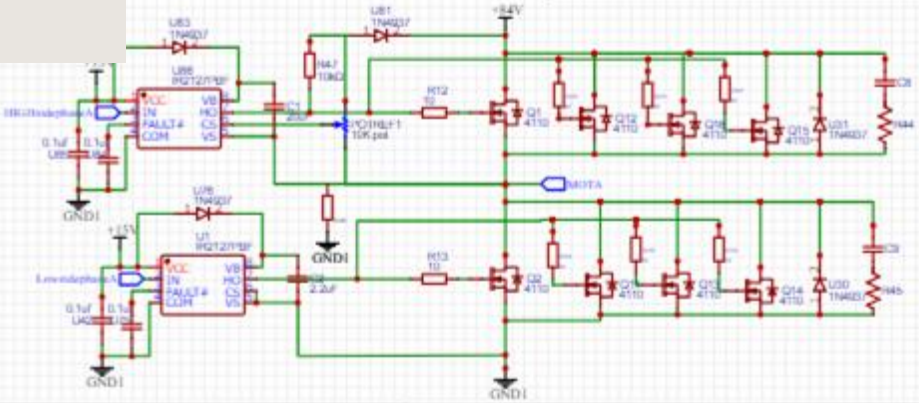
CANBUS



STM CONNECTION



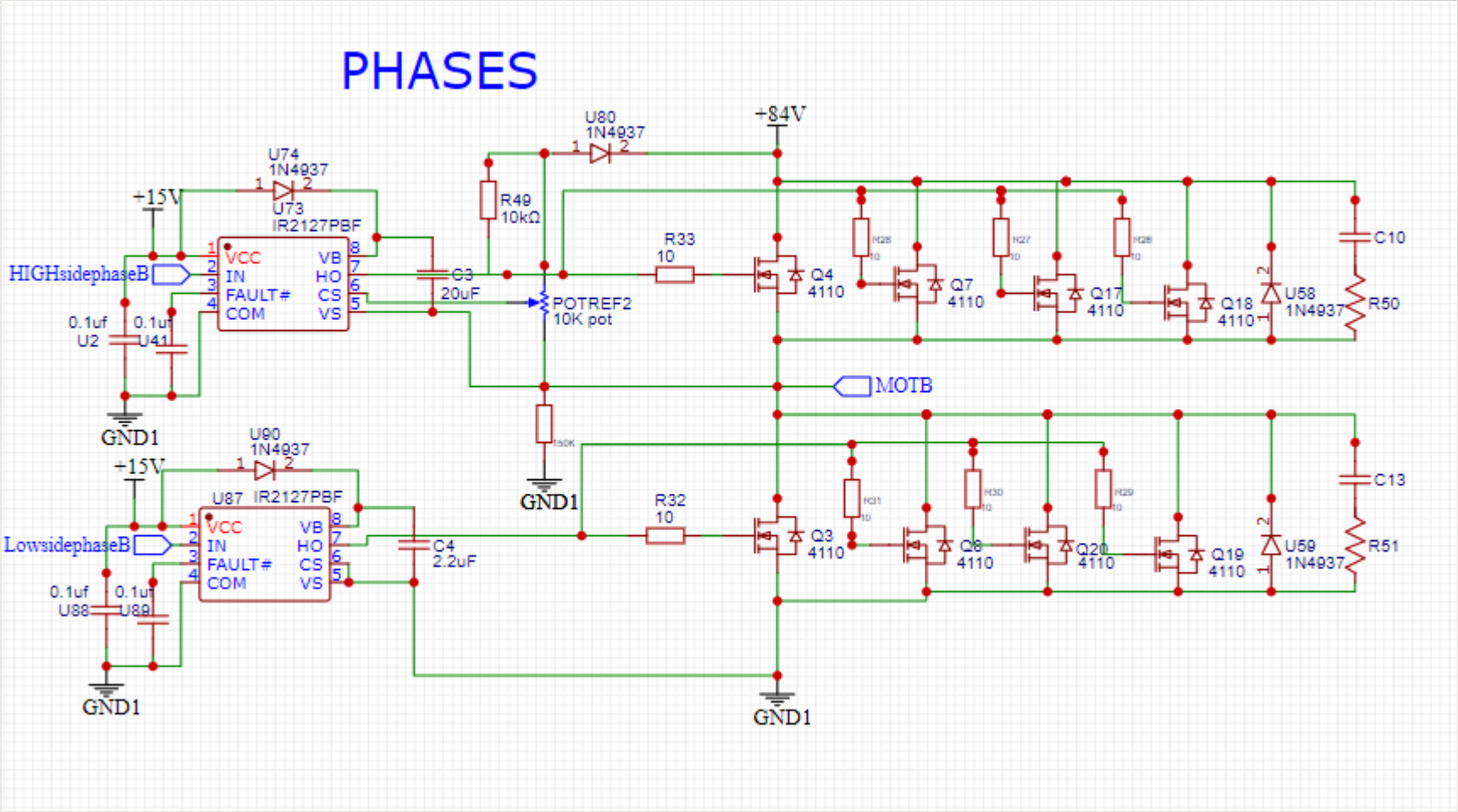




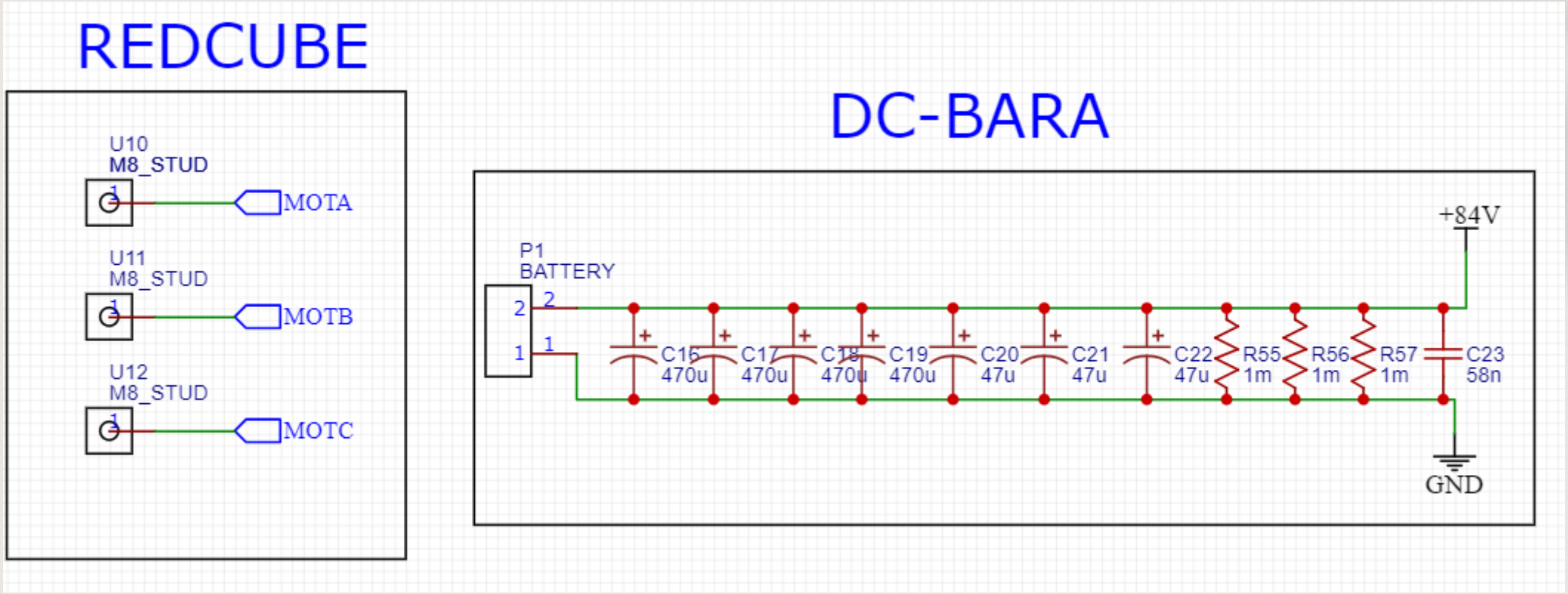
eet_1 +

PCB POWER PART

Gate drivers and mosfets connections.

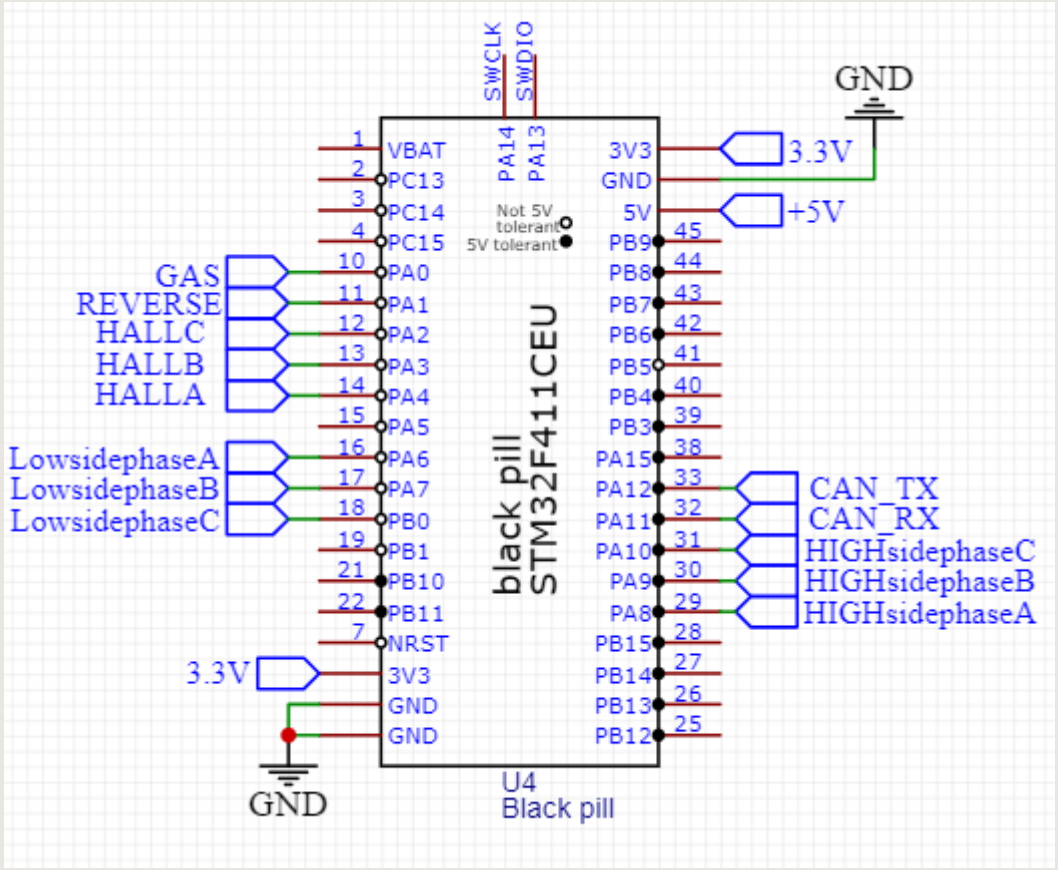


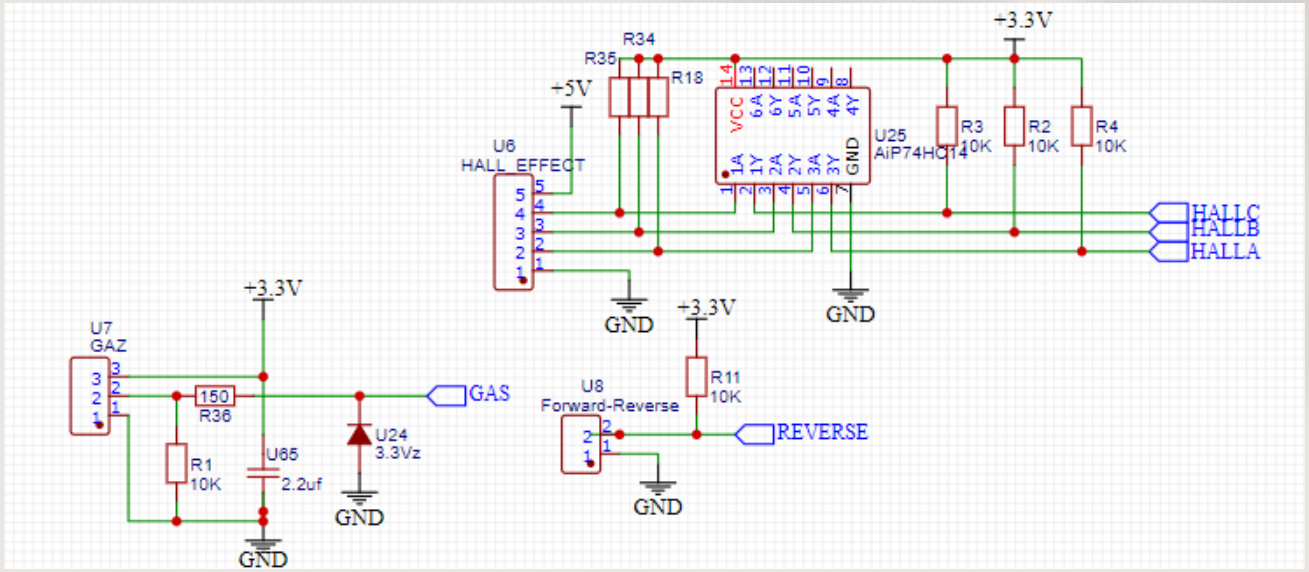
- DC-BARA and RED-CUBE



MICRO PROSSESOR PART

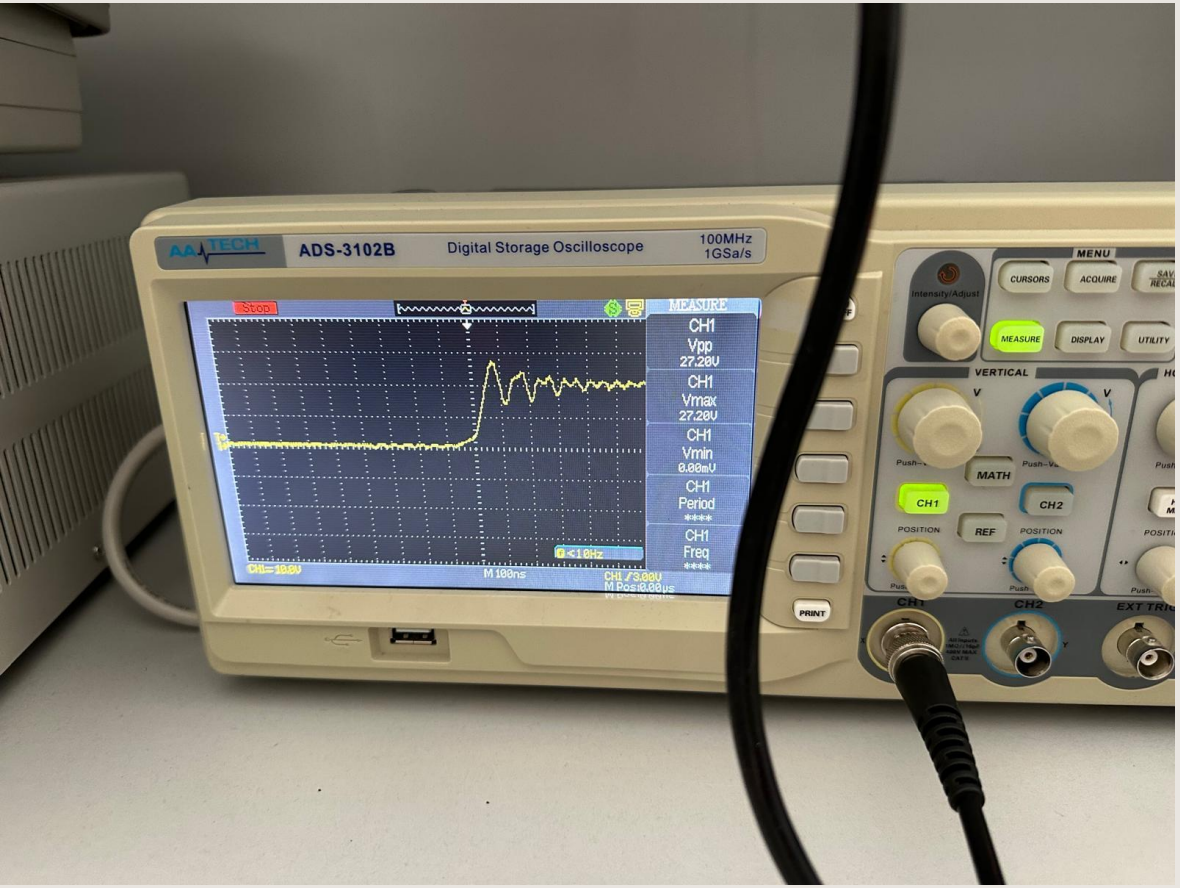
Stm connections



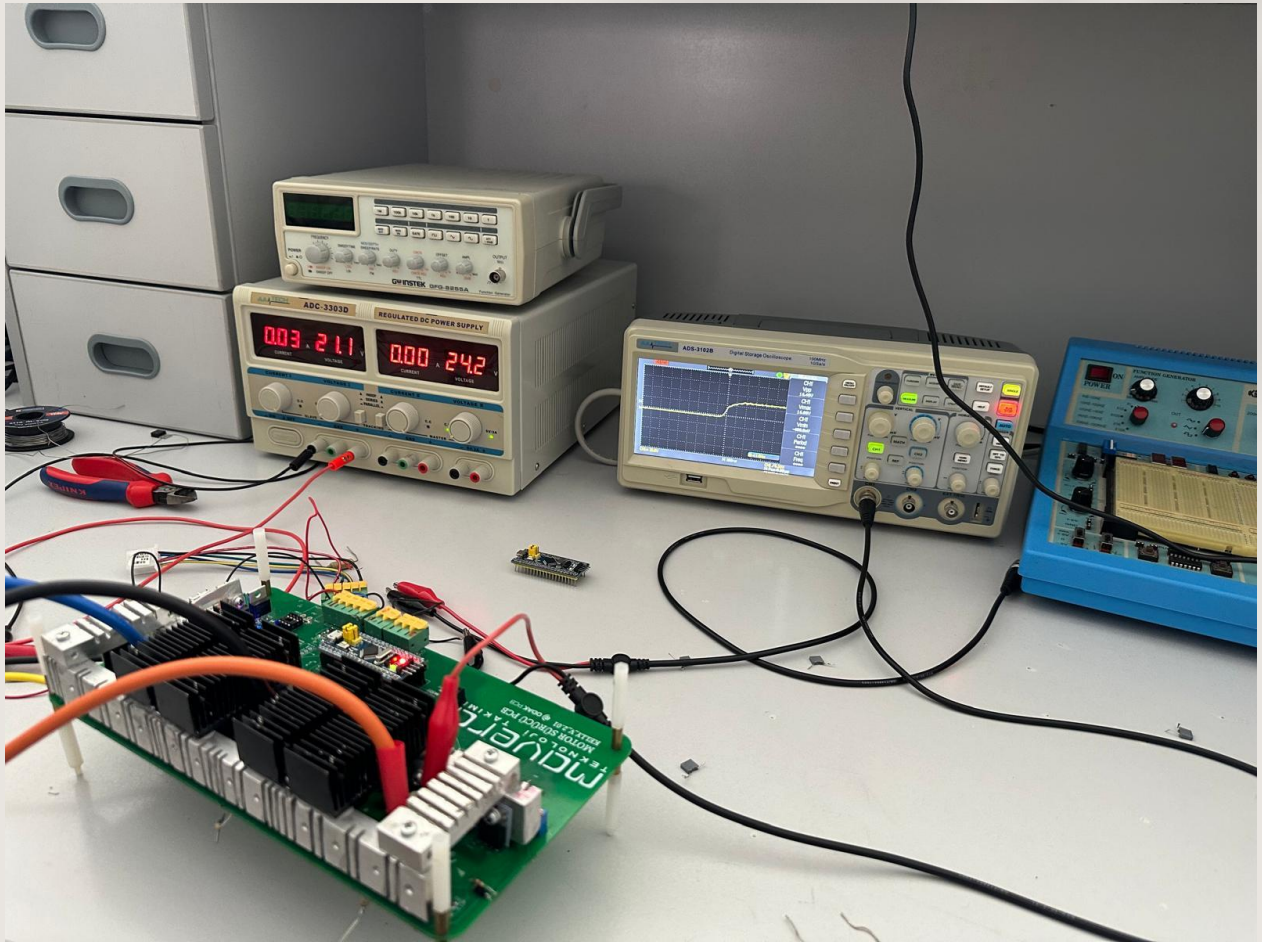


Snubber

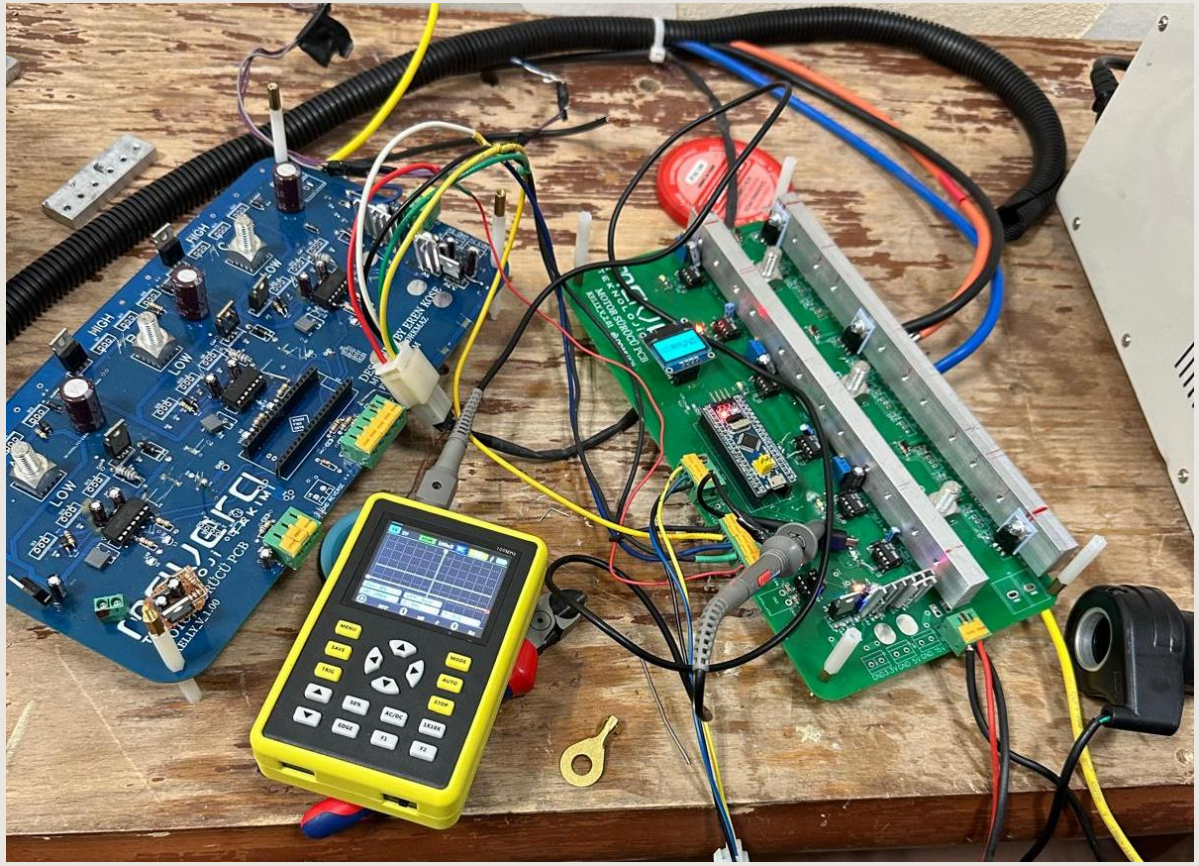
Without snubber

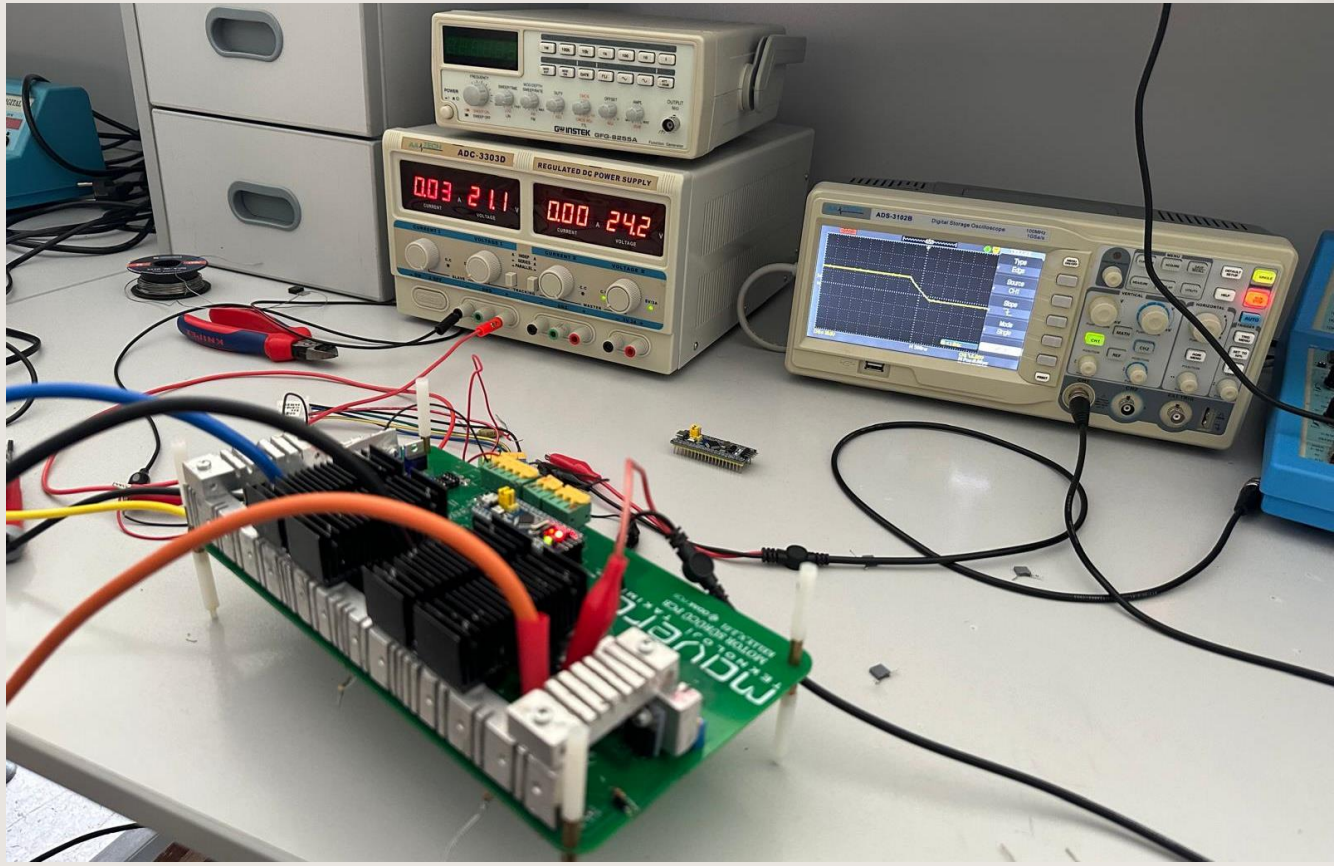


With snubber

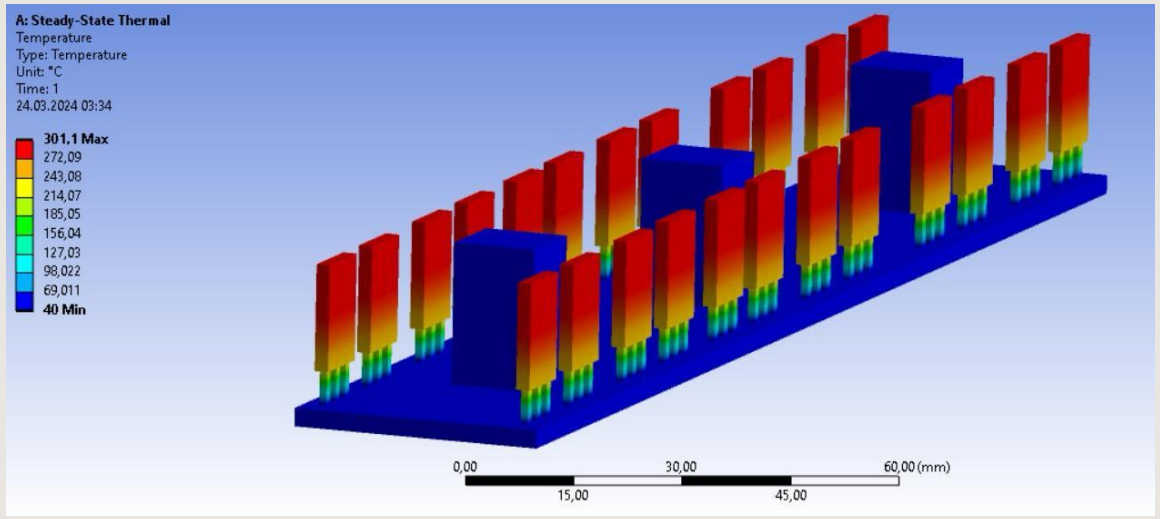


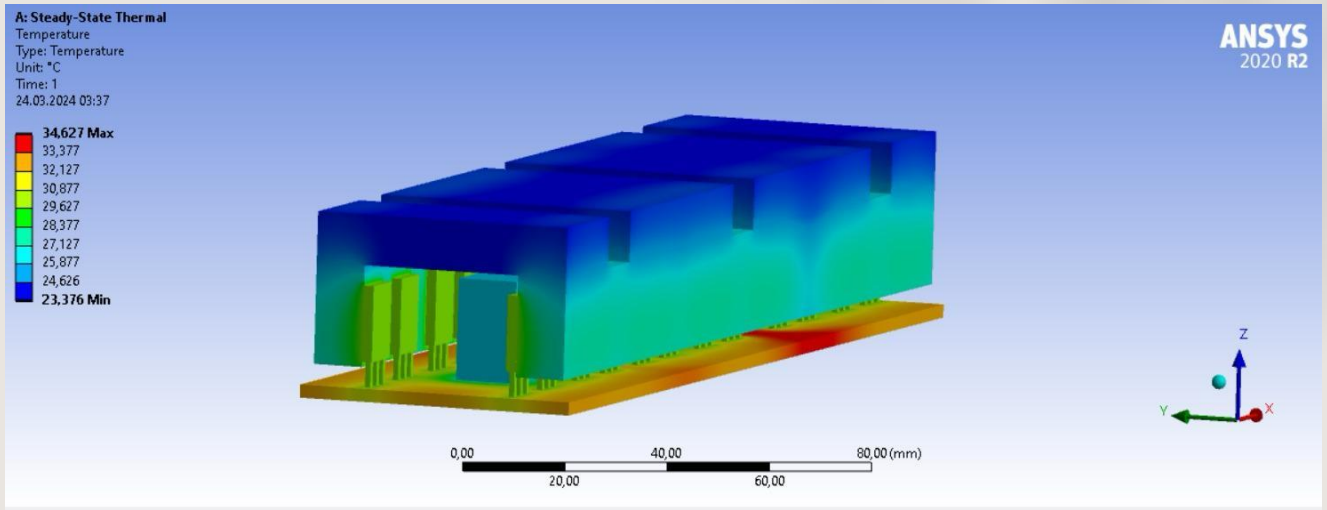
PCB


HIGH
LOW
HIGH
LOW
LOW


ADC-3303D
REGULATED DC POWER SUPPLY
0.03 A 21.1 V
0.00 A 24.2 V
ADS-3102B
Digital Storage Oscilloscope

Cooling system





Software

- Functions work in MOTOR.C and MOTOR.H libraries.
- Timers work in master and slave structure.
- STM is used in two main cutting operations. Engine rotation function Timer 3 global interrupt, systick is used for soft start.

General functions:

```
2  #define __motor_H
3
4  #include "stm32f1xx_hal.h"
5  #ifdef __cplusplus
6  extern "C" {
7  #endif /* __cplusplus */
8  void BLDC_startcond(void);
9  void BLDC_phasecontrol(uint16_t adc);
10 void BLDC_motordefault(void);
11 void BLDC_HallSensorsGetPosition(void);
12 void BLDC_right(uint16_t adc);
13 void BLDC_left(uint16_t adc);
14 int BLDC_adcfilter(int deger);
15 int BLDC_adcfilter_value(void);
16 int BLDC_hallpos_value(void);
17 void BLDC_soft_start(int soft);
18 int BLDC_softstart_value(void);
19 void BLDC_speed(int deger);
20 void  BLDC_rpm_calc(int rpm);
21 int BLDC_rpm_calculated(void);
22 #ifdef __cplusplus
23 }
24 #endif /* __cplusplus */
25
```

Codes in main code

```
449 void HAL_TIM_PeriodElapsedCallback(TIM_HandleTypeDef *htim) {
450     if (htim->Instance == TIM3) {
451
452         filters_value = HAL_ADC_GetValue(&hadc1);
453         if (HAL_GPIO_ReadPin(GPIOA, reverse_Pin) == 1) {
454             BLDC_left(filters_value);
455         } else {
456             BLDC_right(filters_value);
457         }
458         //hall deÄŸerini okuma
459
460         //adc smooth deÄŸer okuma
461
462     }
463 }
464 }
465 void SysTick_Handler(void) {
466     HAL_IncTick();
467     //200ms
468
469     int sysyick_value = HAL_ADC_GetValue(&hadc1);
470     if (sysyick_value  <= 1200) {
471         BLDC_soft_start(0);
472     }
473     if (uwTick % 5 == 0) {
474         ++ctr;
475         BLDC_soft_start(ctr);
476         HAL_GPIO_TogglePin(GPIOC, GPIO_PIN_13);
477         ctr = BLDC_softstart_value();
478     }
479
480 }
```

```
    while (1) {
    /* USER CODE END WHILE */
        int sysyick_value = HAL_ADC_GetValue(&hadc1);
        if(sysyick_value>1170&&sysyick_value<1190){BLDC_startcond();}

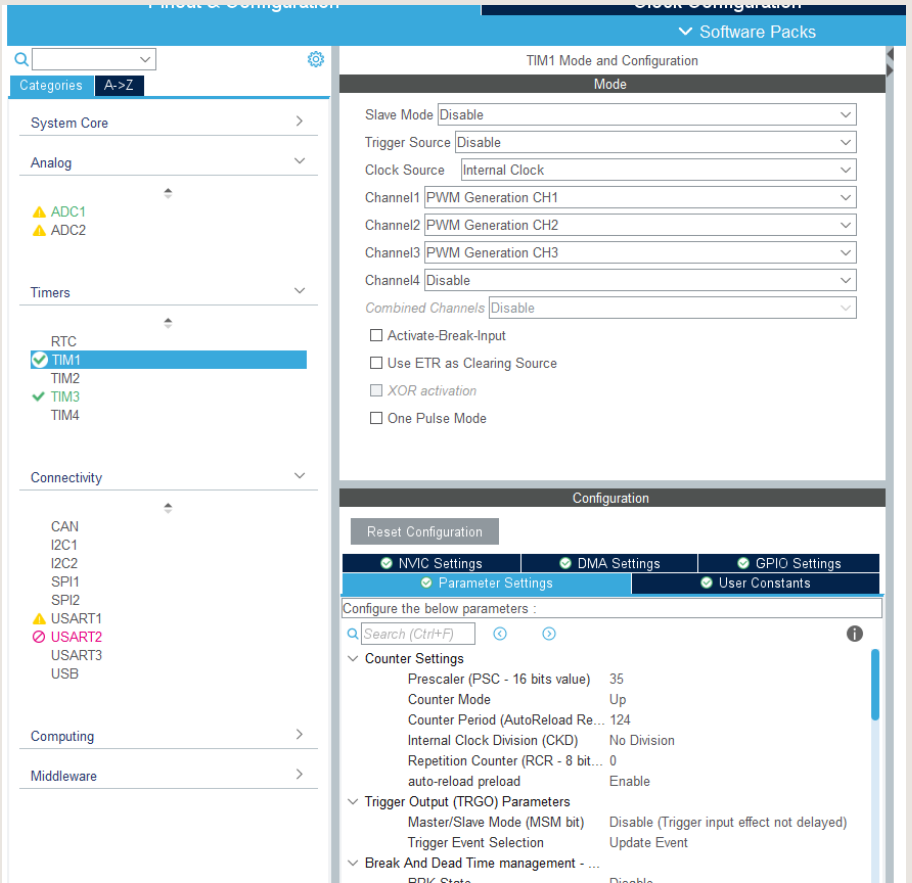
    /* USER CODE BEGIN 3 */

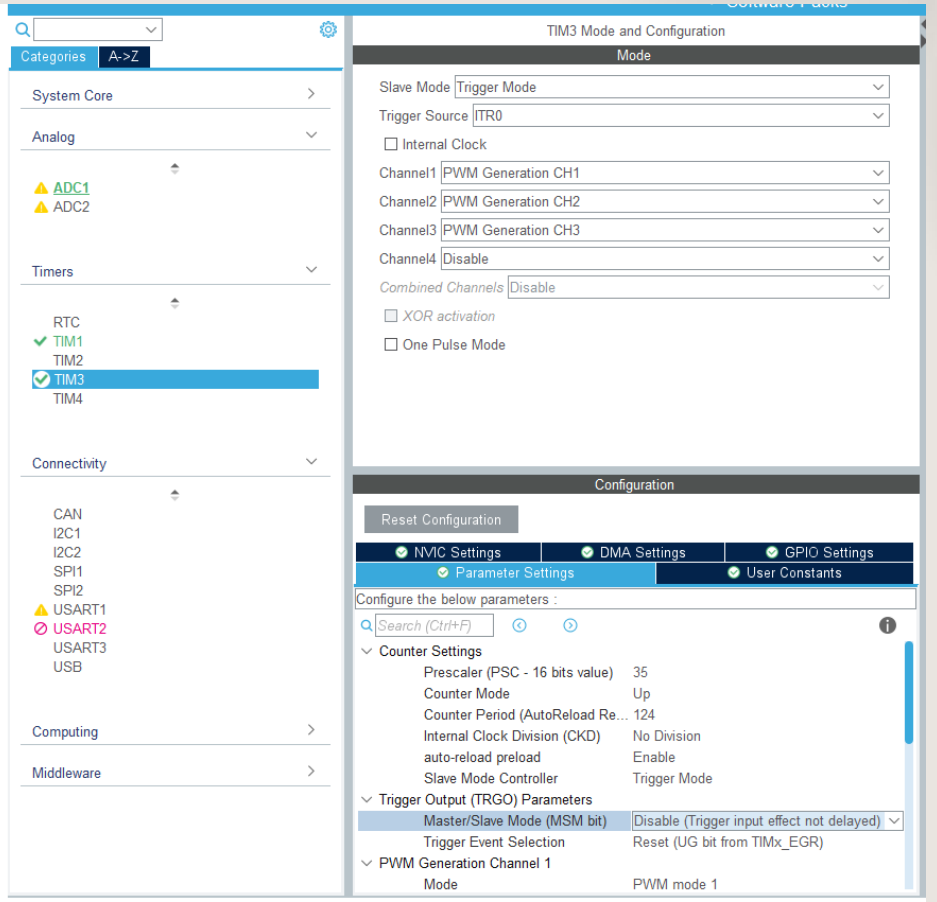
    }
  /* USER CODE END 3 */
}
```

Master slave

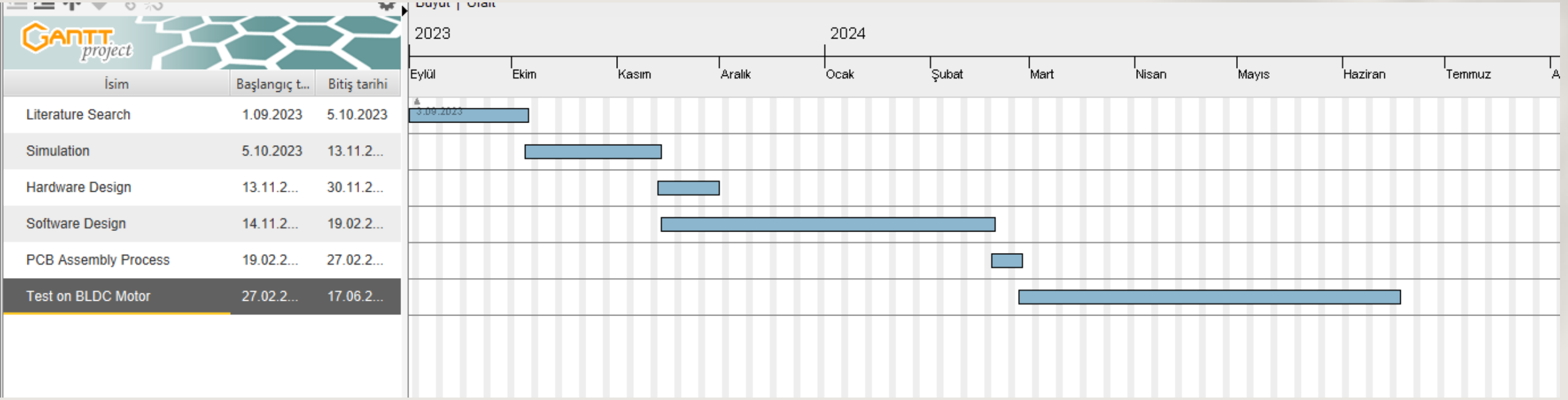
Table 112. TIMx internal trigger connections

Slave TIM	ITR0 (TS = 000)	ITR1 (TS = 001)	ITR2 (TS = 010)	ITR3 (TS = 011)
TIM2	TIM1	TIM8	TIM3	TIM4
TIM3	TIM1	TIM2	TIM5	TIM4
TIM4	TIM1	TIM2	TIM3	TIM8
TIM5	TIM2	TIM3	TIM4	TIM8





TIMING



This project took approximately 2.5 years. but under the final project it took 38 weeks.

CONCLUSION

- Mosfet driving techniques, especially the boostrap method, are understood
- BLDC motors understood.
- Mosfet application methods and results were understood.
- Integrated circuit design and coding were done.
- The information learned will be used in the next card design.

Referances

- [1] "Mcb six step commutation switching," Wikipedia, The Free Encyclopedia. [Online]. Available: https://tr.wikipedia.org/wiki/Dosya:Mcb_six_step_commutation_switching.png.
- [2] H. Özkan, "Blog Yazısı Başlığı," Hamza Özkan Blog, 2022. [Online]. Available: https://blog.hamzaozkan.com.tr/2022/
- [3] P. Sustek, "BLDC Motor Control with Hall Sensors Driven by DSC using TWR-56F8257 and TWR-MC-LV3PH Boards," Systems Application Engineer, Microcontroller Solutions Group, 2022.

Thanks For Listening

- EREN KÖSE
- ERSİN SARI
- ERDİ ŞENTÜRK
- MUSTAFA YAĞCI